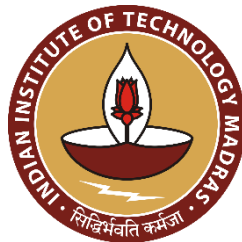


4th International Conference on
Recent Trends in Materials Science & Devices 2026

Design and Finite Element Analysis of Modified Patient Lift for Locomotor Parkinsonism Rehabilitation

Date : 03/04/2026

Abstract ID : 903



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Problem & Motivation

- Affects *gait, balance, posture, and turning ability*.
- The main cause is **axial rigidity**, which leads to **stiffness of the trunk**.
- Reduced trunk mobility leads to:
 - **Poor balance**
 - **Freezing of gait**
 - **Increased risk of falls**

Motivation

- **Trunk movement** plays a critical role in balance and gait initiation.
- Most existing rehabilitation devices focus mainly on **gait training**.
- Very few devices specifically target **trunk rehabilitation**.

Therefore,

*need for a device that focuses on **seated trunk rehabilitation***

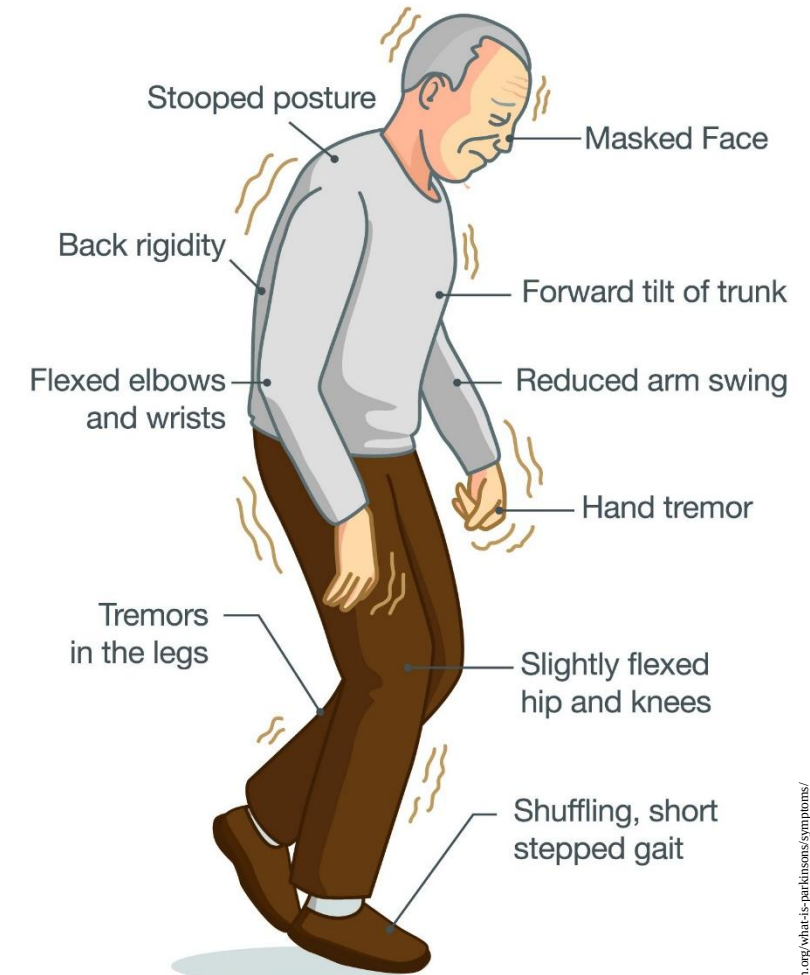


Figure 1: Symptoms of Locomotor Parkinsonism

Objective

To design and develop a **modified patient lift** for trunk rehabilitation in locomotor parkinsonism patients.

- Develop a mechanism for **trunk flexion–extension** using actuator.
- Develop a **manual worm gear mechanism** for lateral bending.
- Enable **active and passive rehabilitation modes**.
- Perform **Finite Element Analysis** to evaluate structural safety.

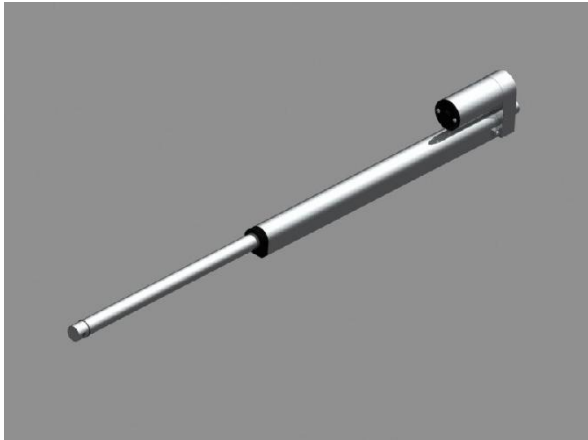


Figure 2: Linear Actuator for automated trunk Flexion & Extension

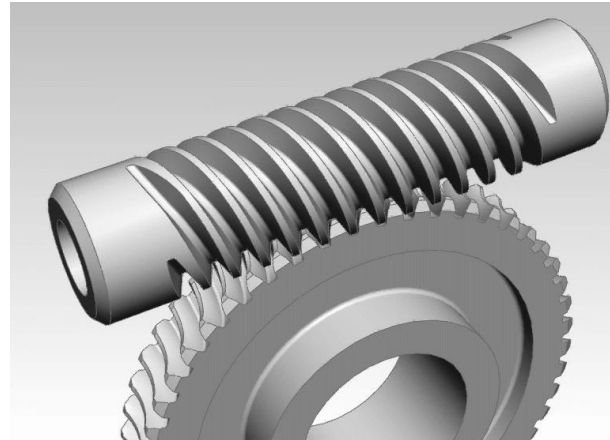


Figure 3: Worm Screw Mechanism for manual lateral bending



Figure 4: Battery Operated Patient Lift

Methodology

- CAD model developed in **PTC Creo**.
- Structural analysis performed in **ANSYS Workbench**.
- Static structural FEA used to evaluate **von Mises stress**.
- Design optimized based on stress distribution.
- Lateral bending mechanism developed as a concept.

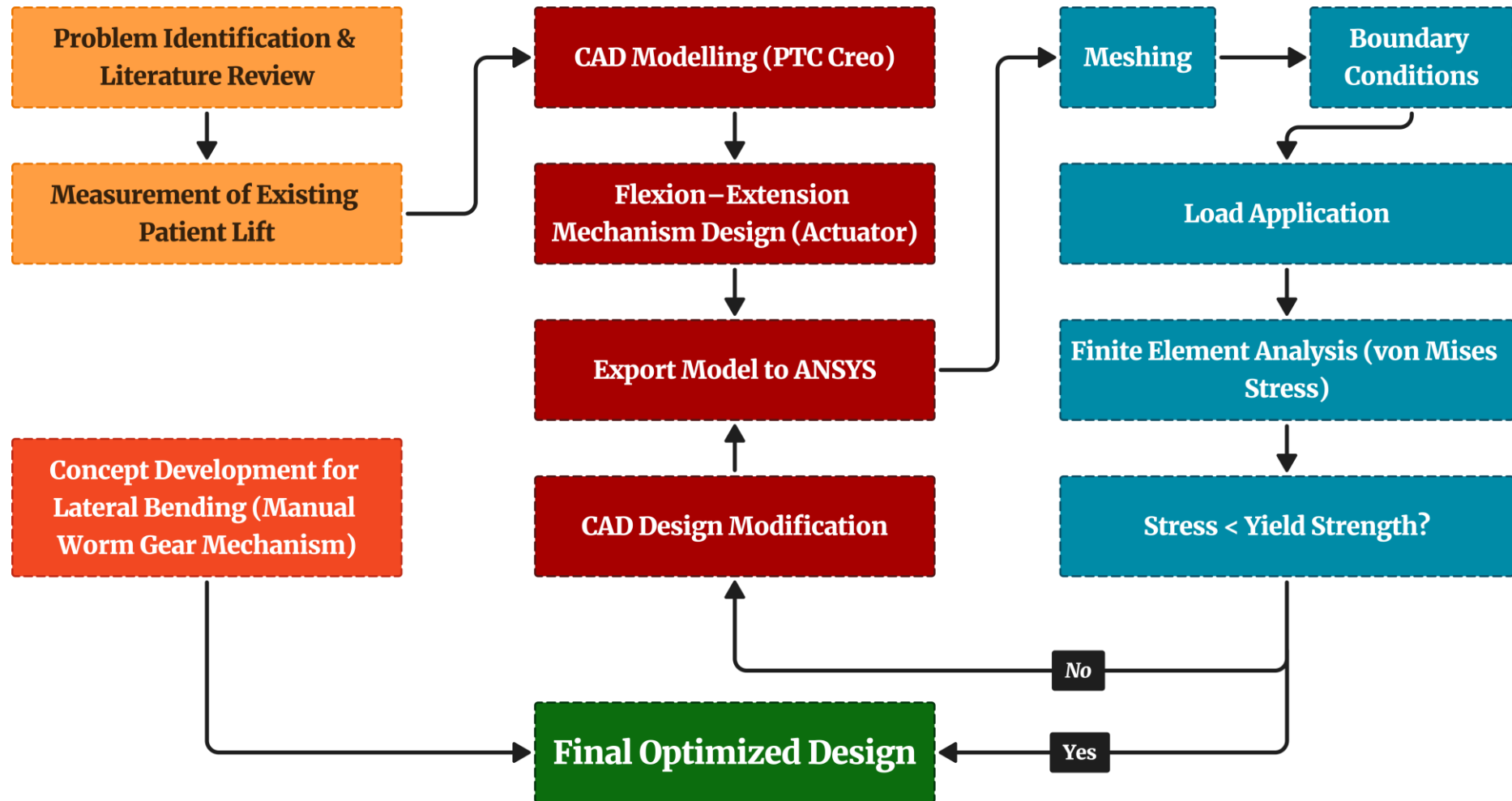


Figure 5: Methodology for the design and finite element analysis of the modified patient lift

Device Design

- The modified patient lift was designed using **PTC Creo**.
- The device consists of:
 - Base frame
 - Seat & Backrest
 - Linear actuator
 - Hinges and support frame
- The actuator enables **trunk flexion and extension**.
- Dimensions were selected based on **anthropometric data**.

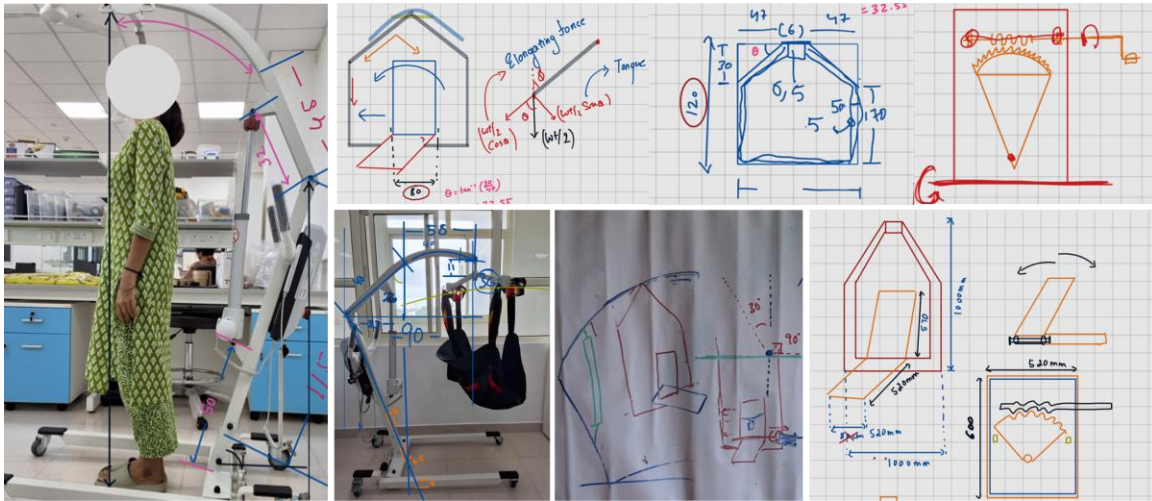


Figure 6: Concept sketch and measurement of the existing device

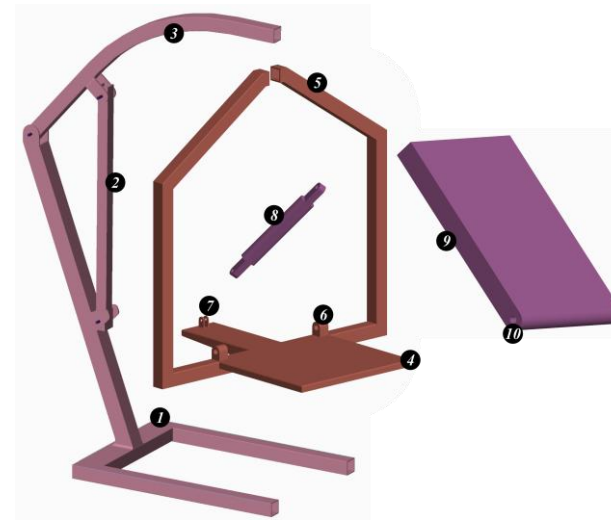


Figure 7: Different parts of modified patient lift

- ### ***Different Parts of the device***
- | | |
|-------------------|---------------------|
| 1- Base Structure | 6- Backrest Holder |
| 2- Base Actuator | 7- Actuator holder |
| 3- Base Arch | 8- Bend Actuator |
| 4- Patient Seat | 9- Patient Backrest |
| 5- Main Frame | 10- Frame hinge |

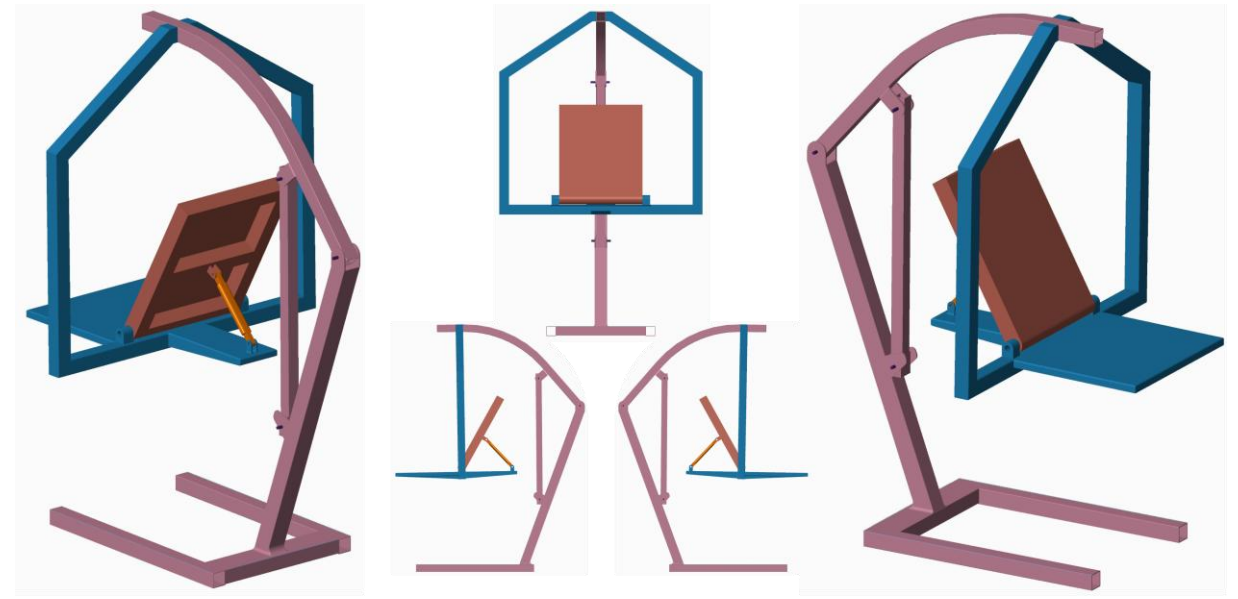


Figure 8: Final CAD model of the modified patient lift.

Finite Element Analysis

- Analysis Type: **Static Structural**
- Structural Material: **Structural Steel**
- Yield Strength: **250 Mpa**
- Mesh Size: **6 mm**
- Loads Applied: **1000 N, 1250 N, 1500 N, 2000 N**
- Boundary Condition: **Base Fixed Support**
- Failure Criterion: **von Mises Stress**

Mesh sensitivity analysis was performed to ensure result accuracy.

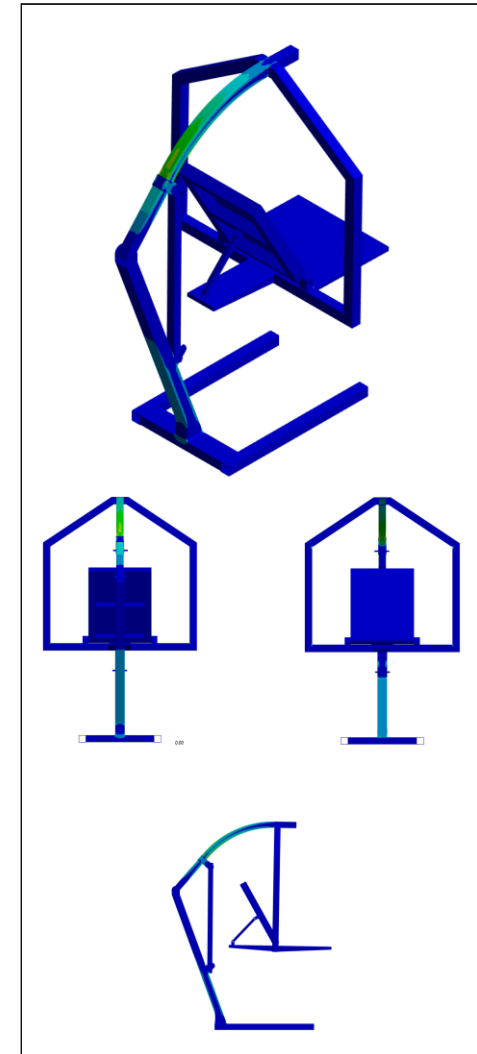
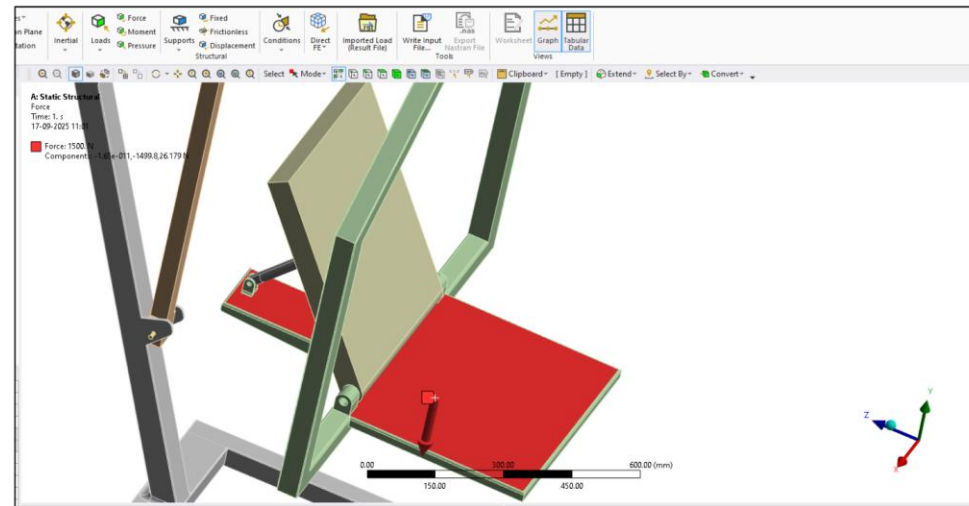
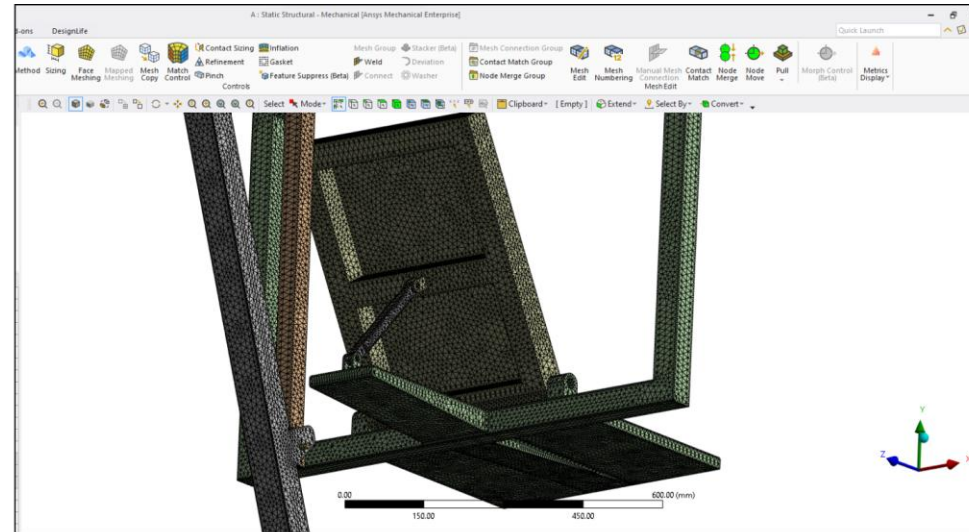
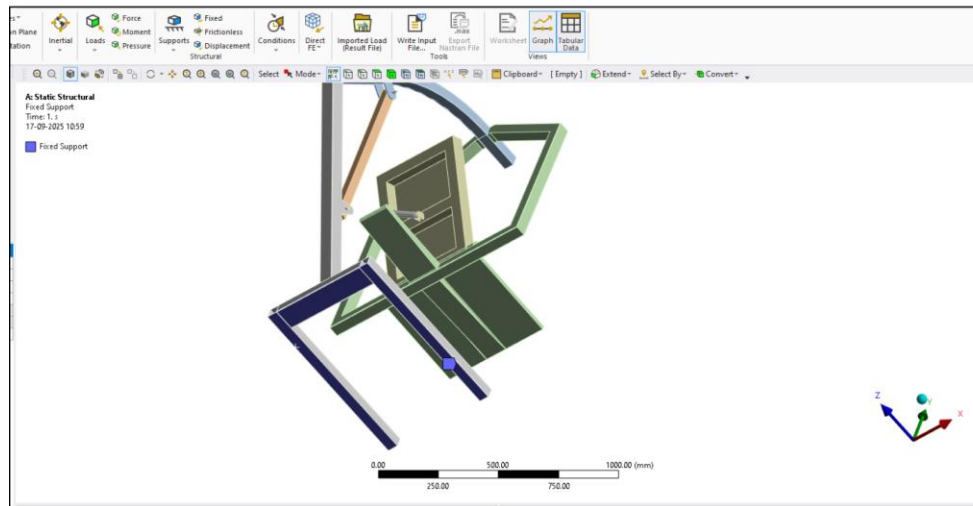


Figure 9: Finite element model showing mesh, boundary conditions and load application

Initial Results – Stress Analysis

- Initial design showed **high stress concentration**
- Maximum stress **exceeded 250 Mpa**
- Failure at:
 - Upright **stand base**
 - Actuator **mounting arch**
- **Design modification required**

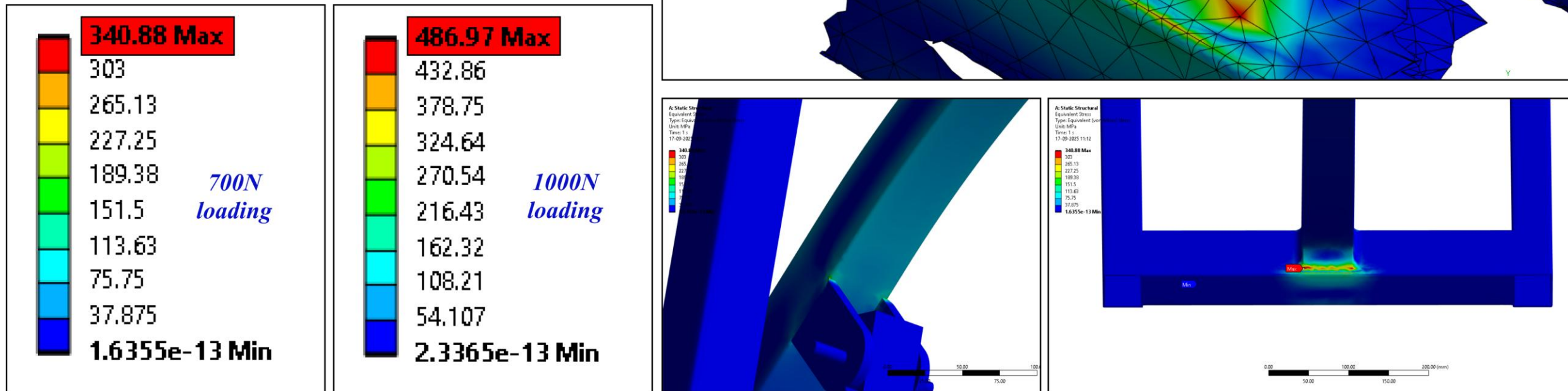


Figure 10: von Mises stress distribution showing stress concentration in the initial design

Final Results – Optimized Design

- **Structural reinforcements** were added at critical locations.
- Stress concentration was **significantly reduced**.
- Maximum stress was **below the yield strength**.
- The structure is safe up to 1200 N load.
- Suitable for patient weight **up to 125 kg**.

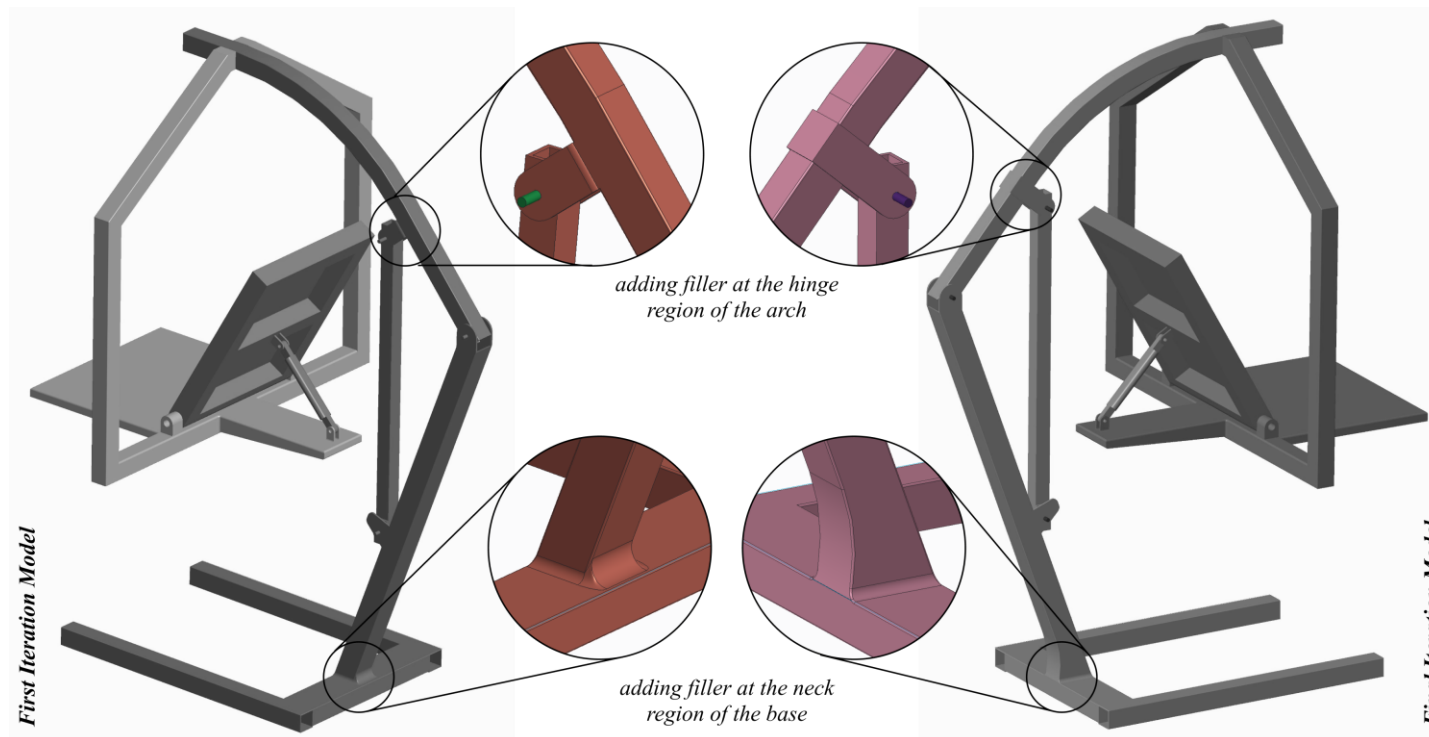


Figure 12: Structural reinforcements done at critical points to strengthen the modified patient lift

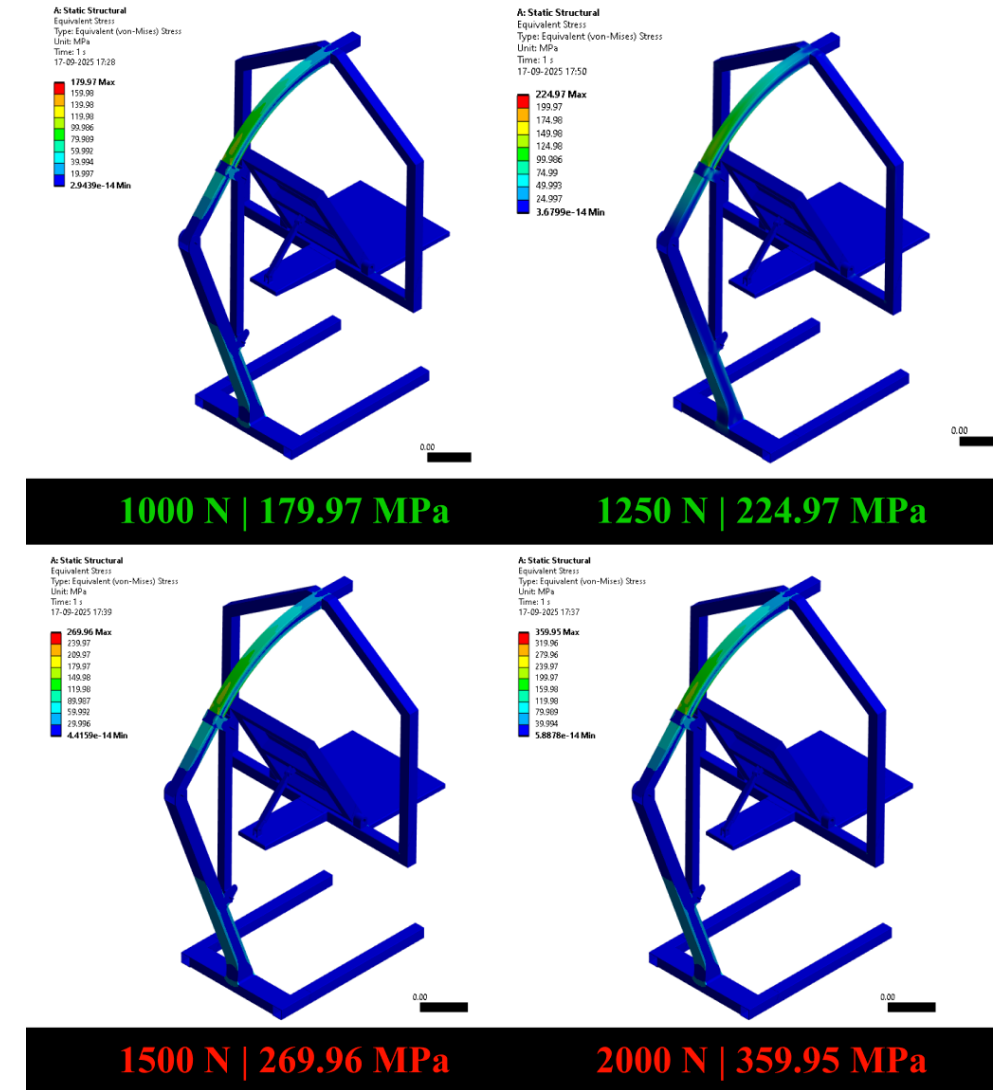


Figure 13: Stress distribution after design modification under different loading conditions

Conclusion

- A modified patient lift for **seated trunk rehabilitation** was successfully designed and developed.
- The device enables **controlled trunk flexion and extension** using a linear actuator mechanism.
- Finite Element Analysis** was performed to evaluate the structural safety under different loading conditions.
- High stress in the initial design was reduced through **design modification and reinforcement**.
- The final optimized design is safe for loads up to **1200 N**, ($\approx$ **125 kg patient weight**).
- The device can be used for **trunk rehabilitation of locomotor parkinsonism patients**.

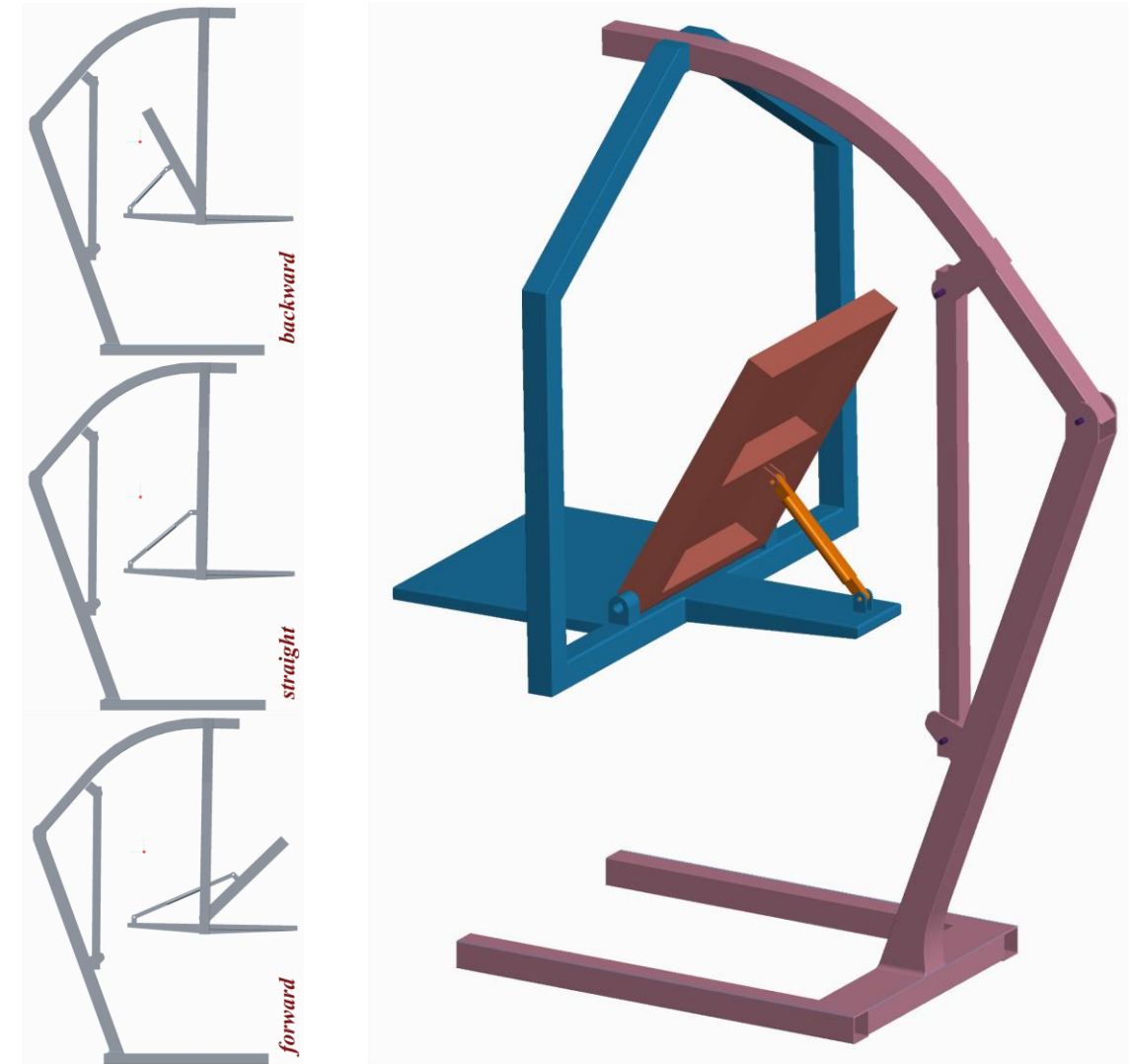


Figure 14: Final Structure of modified patient lift with controlled trunk flexion and extension

Future Work

- Implementation of **lateral bending mechanism** using manual worm gear.
- Addition of **axial rotation mechanism** for trunk movement.
- Fabrication of a **physical prototype**.
- Use of **lightweight materials** such as aluminum.
- **Clinical testing** and validation with patients.

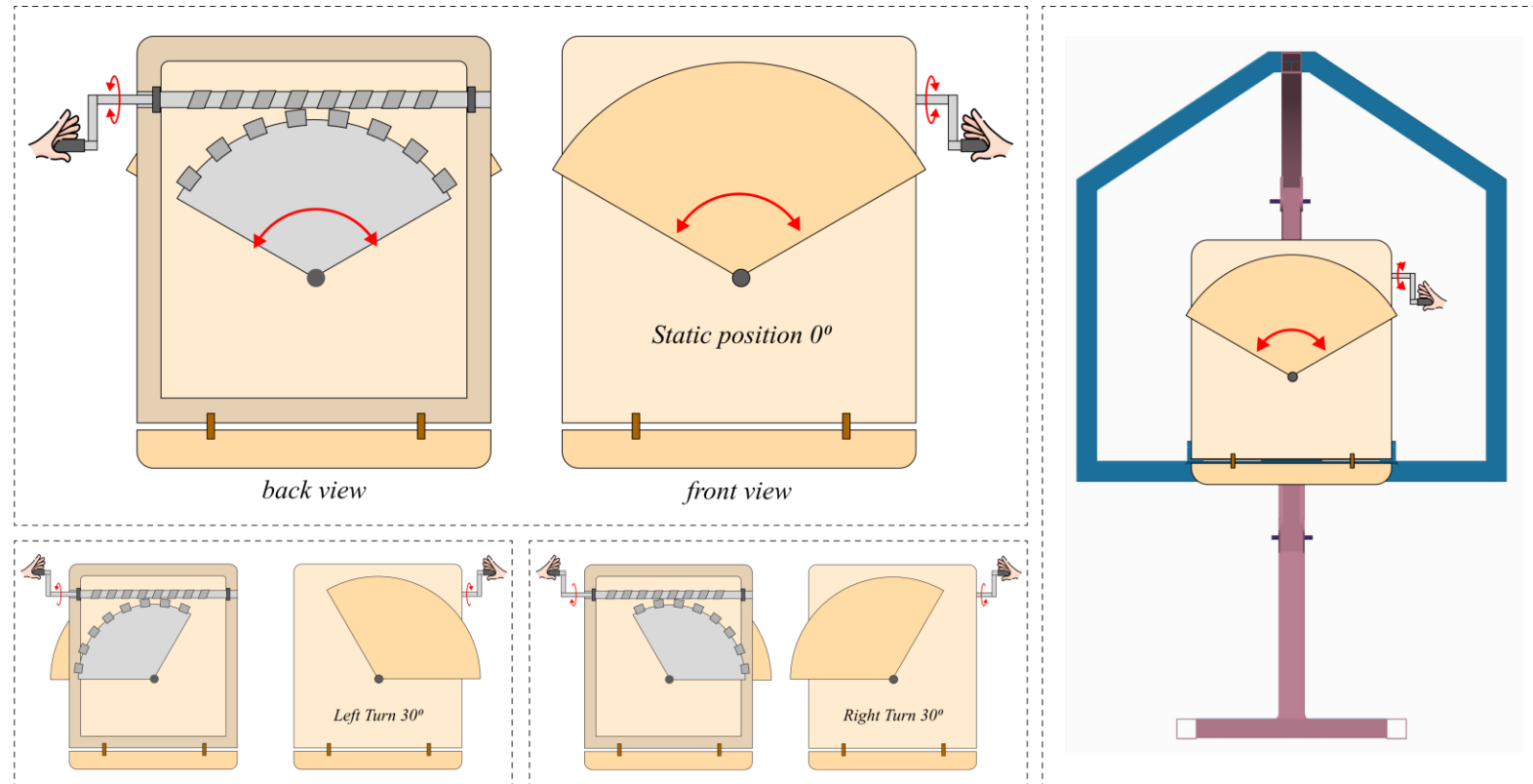


Figure 15: Conceptual design of lateral bending mechanism using manual worm gear

THANK YOU

Questions